

Legal Δ : Enhancing Legal Reasoning in LLMs via Reinforcement Learning with Chain-of-Thought Guided Information Gain

Xin Dai^{1*}, Buqiang Xu^{1*}, Zhenghao Liu^{1†}, Yukun Yan²
Huiyuan Xie², Xiaoyuan Yi³, Shuo Wang², Ge Yu¹

¹Department of Computer Science and Technology, Northeastern University, Shenyang, China

²Department of Computer Science and Technology, Institute for AI, Tsinghua University, Beijing, China

³Microsoft Research Asia, Beijing, China

liuzhenghao@mail.neu.edu.cn

Abstract

Legal Artificial Intelligence (LegalAI) has achieved notable advances in automating judicial decision-making with the support of Large Language Models (LLMs). However, existing legal LLMs still struggle to generate reliable and interpretable reasoning processes. They often default to fast-thinking behavior by producing direct answers without explicit multi-step reasoning, limiting their effectiveness in complex legal scenarios that demand rigorous justification. To address this challenge, we propose **Legal Δ** , a reinforcement learning framework designed to enhance legal reasoning through *chain-of-thought guided information gain*. During training, Legal Δ employs a dual-mode input setup—comprising direct answer and reasoning-augmented modes—and maximizes the information gain between them. This encourages the model to acquire meaningful reasoning patterns rather than generating superficial or redundant explanations. Legal Δ follows a two-stage approach: (1) distilling latent reasoning capabilities from a powerful Large Reasoning Model (LRM), DeepSeek-R1, and (2) refining reasoning quality via differential comparisons, combined with a multidimensional reward mechanism that assesses both structural coherence and legal-domain specificity. Experimental results on multiple legal reasoning tasks demonstrate that Legal Δ outperforms strong baselines in both accuracy and interpretability. It consistently produces more robust and trustworthy legal judgments without relying on labeled preference data. All code and data are available at <https://github.com/NEUIR/LegalDelta>.

1 Introduction

The rapid development of LegalAI systems—referring to artificial intelligence technologies applied to legal practice and judicial processes—is fundamentally reshaping the legal landscape. By automating complex tasks such as legal information extraction (Rabelo et al. 2022), intelligent case retrieval (Zeng et al. 2005; Zhang and Dou 2023), judgment prediction (Liu et al. 2023b), and legal question answering (Louis, van Dijck, and Spanakis 2024), LegalAI significantly enhances the efficiency and accessibility of legal services. These technologies have demonstrated substantial

*These authors contributed equally.

†Corresponding author.

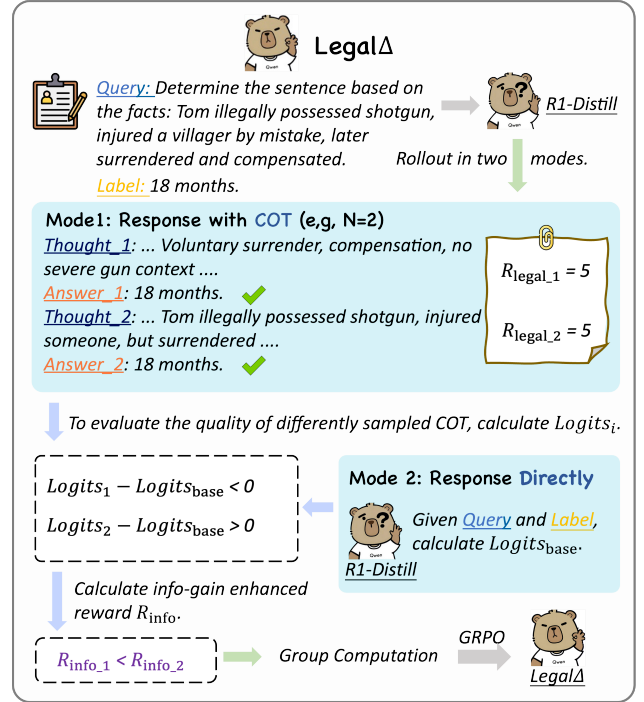


Figure 1: Illustration of Legal Δ . We present how chain-of-thought guided information gain works in RLVR.

value in real-world applications, including contract analysis, legal dispute resolution, and regulatory compliance. Given these promising developments, recent research has increasingly turned to Large Language Models (LLMs) as a foundation for building a more effective legal assistant.

Recently, researchers have increasingly adopted Reinforcement Learning (RL) frameworks such as Group Relative Policy Optimization (GRPO), which enables models to learn from feedback and optimize their behavior through trial-and-error. Large Reasoning Models (LRMs) like QwQ (Team 2025) and DeepSeek-R1 (Guo et al. 2025) have demonstrated strong reasoning capabilities across specialized domains including medicine (Sun et al. 2025), mathematics (Ahn et al. 2024), and programming (El-

Kishky et al. 2025). Recent work has extended these ideas to the legal domain by distilling reasoning traces from powerful RL-trained models such as DeepSeek-V3 (Liu et al. 2024), as in LawGPT (Zhou et al. 2025). However, these methods typically rely on rule-based rewards in Reinforcement Learning with Verifiable Rewards (RLVR) (Mroueh 2025), which focus on output correctness while neglecting the quality and depth of the reasoning process itself. Therefore, how to effectively optimize Chain-of-Thought (COT) within the RL paradigm remains an open question.

To address these challenges, this paper presents Legal Δ , a reinforcement learning framework designed to improve legal reasoning in LLMs by incorporating **Chain-of-Thought guided Information Gain**. Built upon the GRPO paradigm, Legal Δ introduces an information-gain based reward mechanism into the RLVR setting, encouraging the model to explore diverse reasoning trajectories and develop robust reasoning strategies. Specifically, Legal Δ quantifies information gain by leveraging the mathematical equivalence between model logits and Q-values in reinforcement learning. By comparing model responses under direct inference and chain-of-thought prompting, it computes confidence differentials that capture both (i) pointwise mutual information between reasoning steps and final answer correctness, and (ii) global confidence shifts indicating whether reasoning improves decision reliability. This reward is then used to guide policy optimization, pushing the model to discover and reinforce high-quality reasoning behaviors that balance accuracy and prediction confidence.

Our experiments on various LLMs demonstrate that Legal Δ outperforms all baseline models, achieving approximately 10% performance improvements across multiple legal reasoning tasks, while also exhibiting strong generalization to out-of-domain tasks. Further analysis reveals that the information gain module significantly enhances model confidence levels through progressive logits improvements, while also reducing training batch variance for improved training efficiency and stability. Additionally, comprehensive performance quality assessment demonstrates that Legal Δ increases the prominence of legally tokens and generates reasoning chains with lower perplexity scores at the token-level, while consistently outperforming baseline models across different scenarios complexities at the sequence-level.

2 Related Work

Existing legal LLMs mainly rely on Supervised Fine-Tuning (SFT) to help models better understand legal semantics. These methods usually train the models on legal corpora using language modeling objectives. For example, models like LawGPT-zh (Liu et al. 2023a), HanFei (He et al. 2023), and Lawyer LLaMA (Huang et al. 2023) adopt SFT and combine it with the Low-Rank Adaptation (LoRA) strategy to improve training efficiency (Hu et al. 2022; Cui et al. 2023; Deng et al. 2023). One major challenge of SFT lies in the construction of high-quality training data. Recent work (Yue et al. 2023) addresses this by collecting existing publicly available legal datasets, crawling legal texts from the web, and incorporating instruction-tuning datasets (Zhang et al.

2023; Huang et al. 2023; Dai et al. 2023). While these approaches improve performance to some extent, they typically require extensive labeled data and focus on end-to-end optimization over task-specific labels. As a result, SFT-based methods are often susceptible to overfitting and catastrophic forgetting (Luo et al. 2023), which undermines the retention of previously acquired legal knowledge.

To address this challenge, recent studies (Verma et al. 2025; Vara et al. 2025) have explored Reinforcement Learning (RL) to encourage LLMs to engage in broader exploration by specially designed reward mechanisms. Intelligent Legal Assistant (Yao et al. 2025) utilizes the coverage of case elements as the reward signal to guide clarification in legal question answering. SyLeR (Zhang et al. 2025b) addresses the lack of structured reasoning paths in legal question answering by introducing a structure-aware similarity-based reward mechanism that guides LLMs to construct syllogistic reasoning paths. However, its reliance on surface-level matching, making it difficult to capture the complexity and rigor of legal reasoning. Simultaneously, Legal Mathematical Reasoning (Zhang et al. 2025a) employs result-oriented rewards to enhance mathematical reasoning, belonging to the Reinforcement Learning with Verifiable Rewards (RLVR) framework. However, this approach falls short in effectively evaluating intermediate reasoning quality. Therefore, we propose Legal Δ that integrates an information-gain-enhanced reward mechanism into the RLVR framework to provide fine-grained supervision for reasoning quality, enabling LLM to learn more reliable and confidence-aligned legal reasoning strategies.

3 Methodology

In this section, we introduce the preliminaries of legal tasks (Section 3.1). Then we introduce the reinforcement learning based legal optimization method (Section 3.2). Finally, an information-gain-enhanced reward mechanism is proposed, which is based on differential Q-value analysis to assess reasoning quality (Section 3.3).

3.1 Preliminary of LLM Reasoning in the Legal Tasks

Legal reasoning is a complex cognitive process involving fact analysis, application of legal principles, and structured logical deduction. Unlike standard QA tasks, legal reasoning requires multi-step analytical thinking, integration of domain knowledge, and the generation of interpretable argumentation. Our framework focuses on five representative legal tasks that demand such capabilities.

Given a legal query q , our model $\mathcal{M}_{\text{Legal}}$ produces a structured output:

$$q \rightsquigarrow \mathcal{M}_{\text{Legal}} \rightsquigarrow o = (r, a) \quad (1)$$

where r denotes the step-by-step reasoning process and a is the final answer. This structured output ensures both interpretability and reliability, which are crucial in legal applications. We then introduce our reinforcement learning based training strategy in Section 3.2, which equips

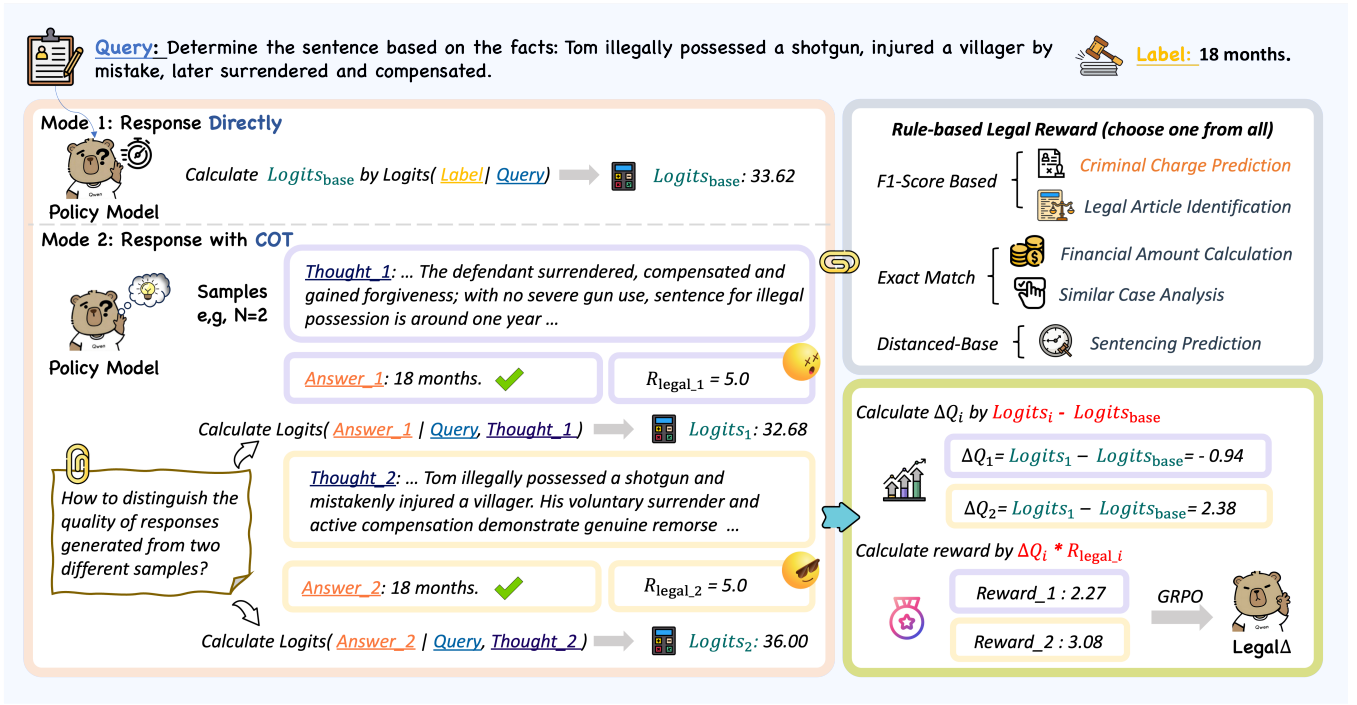


Figure 2: Illustration of Legal Δ .

LLMs with legal reasoning capabilities. Following this, Section 3.3 presents an information-theoretic reward enhancement mechanism that refines R_{legal} into R_{info} via differential Q-value analysis, enabling more nuanced and reasoning-aware reward modeling for reinforcement learning.

3.2 Enhancing the Reasoning Capability of LLMs via Reinforcement Learning

General-domain LLMs often fall short in domain-specific reasoning tasks. Although supervised fine-tuning (SFT) (Gururangan et al. 2020) can adapt models to target domains, it also risks overfitting and catastrophic forgetting (Luo et al. 2023), thereby weakening general reasoning capabilities. To address this, we propose a RL-based training framework. The first stage employs SFT to guide the LLM to imitate the expert-like reasoning behaviors. The second stage applies reinforcement learning to further refine the reasoning ability based on performance feedback.

Supervised Fine-Tuning We construct $\mathcal{M}_{\text{R1-Distill}}$ via SFT using reasoning-enhanced legal data. Specifically, we leverage the high-quality reasoning chains generated by DeepSeek-R1 (Guo et al. 2025) to construct a dataset pairing legal queries with structured and expert-like reasoning steps.

The SFT loss is defined as the standard cross-entropy loss over the full reasoning sequence:

$$\mathcal{L}_{\text{SFT}} = -\mathbb{E}_{(q,r,a) \sim \mathcal{D}_{\text{train}}} [\log P_{\theta}(r, a|q)], \quad (2)$$

where $\mathcal{D}_{\text{train}}$ represents the training dataset, $o = (r, a)$ denotes the complete output containing both reasoning chain

and final answer, and $P_{\theta}(r, a|q)$ represents the probability of generating the complete reasoning output given the query and task instruction.

Reinforcement Learning Building upon the distilled model, we apply Group Relative Policy Optimization (GRPO) (Shao et al. 2024) to further enhance the model’s reasoning ability. Unlike traditional RL approaches that require separate critic models, GRPO compares intra-group outputs to estimate advantages, making it effective for the scenarios where outputs can vary widely in quality.

Within the GRPO framework, our policy model π_{θ} generates multiple reasoning trajectories $\{o_i\}_{i=1}^G$ for each legal query. Then we can use the GRPO training method to optimize LLMs:

$$J_{\text{GRPO}}(\theta) = \mathbb{E}_{q \sim \mathcal{D}, \{o_i\}_{i=1}^G \sim \pi_{\theta_{\text{old}}}(\cdot|q)} \left[\frac{1}{G} \sum_{i=1}^G L_i(\theta) \right] - \beta D_{\text{KL}}(\pi_{\theta_{\text{old}}} || \pi_{\theta}), \quad (3)$$

where $L_i(\theta) = -\log \pi_{\theta}(o_i|q) \cdot A_i^G$, and A_i^G is the group-wise advantage. The KL term penalizes policy shifts to maintain training stability.

The effectiveness of GRPO critically depends on the quality of reward signals. Legal Δ defines a structured reward function as:

$$R(o_i) = R_{\text{format}}(o_i) + R_{\text{legal}}(a_i), \quad (4)$$

where R_{format} ensures adherence to the structured output format, and R_{legal} evaluates the correctness of the legal reasoning process and conclusion.

Format Reward ($R_{\text{format}}(o_i)$): This component ensures outputs adhere to a structured format, with `<reasoning>` and `<answer>` tags separating the reasoning chain from the final decision. This format facilitates reward computation of both the reasoning process and the legal outcome.

Legal Reward ($R_{\text{legal}}(a_i)$): This component evaluates the correctness and legality of the final answer a_i using fine-grained, task-specific criteria. These legal criteria make the outcomes more verifiable and thus suitable for guiding GRPO-based training. Further details on the design of the legal reward are provided in Appendix A.3.

3.3 Information-Gain Enhanced Legal Reward Modeling

To move beyond terminal answer evaluation, we introduce an information-theoretic enhancement to reward design. Specifically, we replace the legal reward R_{legal} component with R_{info} for reward modeling:

$$R(o_i) = R_{\text{format}}(o_i) + R_{\text{info}}(a_i). \quad (5)$$

The reward R_{info} incorporates reasoning confidence through differential Q-value analysis to better evaluate the semantic contribution of reasoning processes in legal analysis:

$$R_{\text{info}}(a_i) = R_{\text{legal}}(a_i) \times \sigma(\Delta Q \cdot T), \quad (6)$$

where $\sigma(\cdot)$ is the sigmoid function providing bounded amplification within (0,1) with monotonic enhancement properties, ΔQ denotes the information quality differential, and T is the temperature parameter that controls the sensitivity of reward modulation for fine-grained control over training dynamics. The multiplicative structure preserves the primacy of correctness while enabling adaptive enhancement based on information content.

Recent work by Li et al. (Li et al. 2025) proves that logits from next-token prediction training are mathematically equivalent to Q-functions in offline inverse reinforcement learning (see detailed proof in Appendix A.4):

$$Q(q, a) = \text{logit}_{\theta}(a|q), \quad (7)$$

where $Q(q, a)$ quantifies the expected value of generating answer a given query q , and $\text{logit}_{\theta}(a|q)$ represents the average unnormalized log-probability scores across all tokens in answer a when generated autoregressively given context q . This theoretical foundation enables us to extract Q-values directly from any pre-trained language model and design a differential mechanism to quantify reasoning quality. For answer a , the Q-value gain given by reasoning process r is expressed as:

$$\Delta Q(r) = \text{logit}_{\theta}(a|q, r) - \text{logit}_{\theta}(a|q). \quad (8)$$

This formulation measures the semantic shift in the model’s internal representation space induced by the reasoning process r , providing a quantitative measure of how much the reasoning contributes to the model’s confidence in generating the correct answer. We show our reasoning mode and direct mode prompt templates respectively in Appendix A.6.

From an information-theoretic perspective, given that logits follow the softmax probability distribution:

$$p_i(a) = \frac{e^{\text{logits}_i(a)}}{Z_i}, \quad (9)$$

where $Z_i = \sum_{a'} e^{\text{logits}_i(a')}$, a' is used to traverse all possible answer tokens. Through algebraic manipulation of the logit difference, we obtain:

$$\Delta Q(r) = \log \frac{p(a|q, r)}{p(a|q)} + \log \frac{Z_{q, r}}{Z_q}. \quad (10)$$

This decomposition reveals the information-theoretic interpretation of our approach.

The first term $\log \frac{p(a|q, r)}{p(a|q)}$ represents the pointwise mutual information (PMI) between the answer token a and the reasoning process r , measuring the correlation between the reasoning chain and the specific answer: a positive value indicates that the reasoning process makes this answer more likely, a negative value means the reasoning actually decreases the probability of this answer, and zero implies no correlation.

The second term $\log \frac{Z_{q, r}}{Z_q}$ serves as a global confidence regularizer that captures whether the reasoning process r enhances the model’s overall decisiveness: when positive, it indicates that r has sharpened the probability distribution, making the model more confident overall; when negative, it suggests the reasoning has introduced confusion. Within the GRPO framework, since rewards are utilized for comparative evaluation across multiple generated reasoning trajectories, the global normalization constant remains invariant across all candidates for the same query, allowing us to focus on the information-theoretic signal that truly differentiates reasoning quality.

4 Experimental Methodology

In this section, we describe the datasets, baselines, evaluation metrics, and implementation details in our experiments.

4.1 Datasets

In our experiments, we select training data from portions of the CAIL2018 (Xiao et al. 2018), JEC-QA (Zhong et al. 2019), and LAIC2021 (Zhong et al. 2019) datasets, which encompass diverse legal reasoning scenarios across multiple domains. We first use 450 high-quality reasoning samples distilled from DeepSeek-R1 to establish foundational legal reasoning capabilities. Subsequently, for GRPO training of Legal Δ , we collect 6,981 samples for the training set and 563 samples for the validation set across all tasks in our experiments. We select knowledge-intensive legal reasoning tasks for a comprehensive assessment, including legal article prediction (SPP-F), criminal charge prediction (CCP), sentencing prediction (SLP), similar case analysis (CAS) and financial amount calculation (CAC), which demand sophisticated reasoning over legal facts and regulations. To assess the generalization capability of our approach, we additionally evaluate performance on the out-of-domain DiscLaw benchmark, which contains various legal professional examinations distinct from our training domains. Detailed dataset statistics can be found in the Appendix A.1.

4.2 Baselines

In our experiments, we compare Legal Δ with existing law models: Fuzi-7B (Deng et al. 2023), HanFei-7B (He et al.

Method	In-Domain								Out-of-Domain					
	SPP-F		CCP		SLP	CAS	CAC	Avg.	PAE	NJE	PFE	UNGEE	LBK	Avg.
	<i>F1</i>	<i>JAC</i>	<i>F1</i>	<i>JAC</i>	<i>Acc</i>	<i>Acc</i>	<i>Acc</i>		<i>Acc</i>	<i>Acc</i>	<i>Acc</i>	<i>Acc</i>	<i>Acc</i>	
<i>Law Model</i>														
LexiLaw-6B	13.15	13.14	39.99	35.63	8.30	20.80	35.80	23.83	23.73	20.11	31.76	31.56	40.36	29.50
Hanfei-7B	0.26	2.65	30.96	27.38	17.10	23.60	39.40	20.19	21.19	14.53	31.76	24.38	29.09	24.19
Fuzi-7B	25.19	27.51	55.93	49.03	15.60	0.70	47.20	31.59	26.27	24.95	33.53	28.13	40.73	30.72
DiscLaw-13B	38.75	48.16	52.87	48.03	41.10	0.00	16.80	35.10	40.68	42.09	57.06	50.94	54.91	49.14
ChatLaw-13B	33.28	32.55	27.90	24.54	11.60	28.80	41.40	28.58	31.36	27.56	42.35	35.62	41.09	35.60
<i>Qwen2.5-3B-Instruct</i>														
Zero-shot	62.37	65.52	50.10	44.85	55.10	34.80	55.80	52.65	47.46	46.55	74.71	70.94	74.18	62.77
SFT (DiscLaw)	68.09	64.25	43.24	38.38	45.50	39.20	46.00	49.24	38.98	43.20	61.76	53.75	57.82	51.10
DPO	64.93	65.02	50.42	43.33	59.50	52.40	56.40	56.00	45.76	46.55	74.12	71.88	73.45	62.35
R1-Distill	67.42	67.74	48.35	43.67	48.20	36.60	57.40	52.77	53.39	47.86	76.47	73.13	75.27	65.22
LegalΔ	79.13	76.20	58.59	54.15	61.90	57.40	63.20	64.37	55.08	50.09	81.76	75.63	76.00	67.71
<i>Qwen2.5-7B-Instruct</i>														
Zero-shot	77.10	75.46	53.81	48.43	48.30	59.80	66.80	61.39	66.94	58.84	90.00	84.37	82.90	76.61
SFT (DiscLaw)	72.71	67.72	41.79	37.56	51.60	38.40	55.20	52.14	47.46	42.09	68.23	55.00	62.55	55.07
DPO	77.23	73.56	55.28	50.05	68.40	56.60	64.40	63.65	66.10	60.34	90.59	85.94	84.73	77.54
R1-Distill	80.14	78.12	54.73	49.92	50.20	55.60	69.20	62.56	66.10	62.75	91.17	82.81	84.72	77.51
LegalΔ	83.14	79.68	68.53	62.90	64.80	64.80	75.80	71.38	67.80	64.43	92.94	82.19	85.09	78.49
<i>Qwen2.5-14B-Instruct</i>														
Zero-shot	84.18	79.66	58.50	52.10	42.60	55.40	76.80	64.18	74.58	65.92	90.59	86.56	87.27	80.98
SFT (DiscLaw)	81.36	77.13	55.53	49.84	37.90	52.20	68.40	60.34	42.37	44.32	62.94	51.87	58.55	52.01
DPO	82.21	77.00	59.65	53.00	58.00	69.20	79.40	68.35	67.80	59.22	91.76	85.63	84.73	77.83
R1-Distill	84.73	80.64	59.93	53.02	45.10	69.80	78.20	67.35	72.88	66.48	92.35	85.31	87.27	80.86
LegalΔ	85.56	81.10	62.79	56.95	67.10	72.40	79.40	72.19	77.12	68.90	92.35	87.19	88.73	82.86

Table 1: Overall Performance.

2023), LexiLaw-6B (Li et al. 2024), ChatLaw-13B (Cui et al. 2023), and DiscLaw-13B (Yue et al. 2024). These law models vary in parameter scales and capabilities, all trained using various training methods on diverse legal datasets. To ensure fair and rigorous evaluation, we reimplemented a representative method (DiscLaw) based on the same Qwen2.5 series base models as our approach. Additionally, we compare with four baseline models: Zero-shot, R1-distill, DPO, and SFT (DiscLaw) for comprehensive evaluation. By evaluating LLMs across different scales, we can systematically assess the generalization ability of various training strategies.

4.3 Evaluation Metrics

We design comprehensive evaluation metrics for different task types following Zhou et al. (2025); Yue et al. (2023). For in-domain evaluation, we employ F1 and Jaccard (JAC) scores for SPP-F and CCP tasks, which assess the quality of multi-label classification. For SLP, CAS, and CAC tasks, we use accuracy scores. For out-of-domain evaluation, we assess generalization capability using accuracy scores across five tasks from the DiscLaw benchmark: PAE, NJE, PFE, UNGEE, and LBK.

4.4 Implementation Details

In our experiments, we adopt the Qwen2.5-Instruct series due to its superior reflective capabilities and cognitively informed reasoning behavior (Gandhi et al. 2025), including

the 3B, 7B, and 14B parameter models (Qwen et al. 2025), as backbone language models to ensure a comprehensive evaluation across different model scales. During training, we use a batch size of 2 per GPU and a learning rate of $5e-5$. For LoRA fine-tuning, we use a rank of 32. We use nucleus sampling with temperature of 0.9 and generate $G = 6$ samples per prompt. For information-gain, we set a temperature of $T = 0.2$. More training implementation details of Legal Δ are presented in Appendix A.2.

5 Evaluation Result

In this section, we first evaluate the performance of various legal LLMs on legal reasoning tasks, demonstrating Legal Δ ’s effectiveness across different model scales and its generalization to out-of-domain scenarios. We then analyze the effectiveness of our information-gain-enhanced reward modeling by examining logits variations during training and the correlation between information gain and reward values. Following this, we conduct a comprehensive performance quality assessment from both token-level and sequence-level perspectives: at the token level, we analyze legal token prominence progression and assess reasoning chain quality through perplexity evaluation; at the sequence level, we examine performance improvements across different scenarios complexities. Additionally, we provide some case studies in Appendix A.5.

5.1 Overall Performance

This section evaluates the legal reasoning capabilities of Legal Δ on both in-domain and out-of-domain tasks.

As shown in Table 1, Legal Δ achieves consistently superior performance across different model scales and evaluation settings. On in-domain legal reasoning benchmarks, Legal Δ outperforms all baselines, including Zero-shot, SFT (DiscLaw), DPO, and R1-Distill, with an average improvement of approximately 10% across Qwen2.5 models of varying capacities. While DPO benefits from preference optimization and R1-Distill leverages high-quality reasoning traces, Legal Δ outperforms both, especially on complex reasoning tasks such as SLP and CAS.

Furthermore, the advantages of Legal Δ extend to out-of-domain scenarios, where it maintains strong performance across diverse evaluation benchmarks. This highlights the generalization capability of our approach, which dynamically optimizes reasoning processes and better captures nuanced legal logic by using the information-gain enhanced reward modeling method.

5.2 Ablation Study

To systematically assess the contribution of each core component in the Legal Δ framework, we conduct a series of ablation experiments. The results are summarized in Table 2.

Our reinforcement learning based training strategy brings substantial performance gains over the vanilla baseline. Removing the GRPO component (*w/o GRPO*) leads to a marked drop in performance, highlighting the critical role of reinforcement learning in enabling the legal reasoning capability of LLMs. This demonstrates that GRPO’s intra-group comparison mechanism effectively exploits quality differences among reasoning trajectories, making it essential for optimizing legal reasoning capabilities. Similarly, the *w/o SFT* setting confirms that both SFT and RL stages are jointly necessary for achieving optimal performance.

Among all components, the information-gain-enhanced reward mechanism contributes most significantly to performance improvements. To verify this, we evaluate a variant that excludes information gain and instead relies solely on traditional reward signals (*w/o Info-Gain*). The resulting performance degradation confirms the value of incorporating differential Q-value analysis, which enables fine-grained differentiation of reasoning quality. The gap between this variant and the full Legal Δ model underscores the effectiveness of our reward enhancement in capturing the subtle characteristics of high-quality legal reasoning.

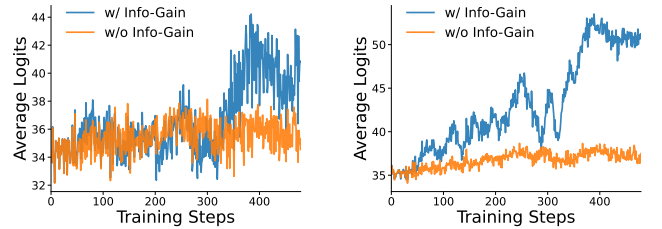
5.3 The Effectiveness of Information-Gain Enhanced Reward Modeling

In this section, we analyze the effectiveness of the information gain module during training through two perspectives: (1) its influence on model confidence via logits analysis, and (2) its impact on training efficiency, measured by batch reward variance.

In deep learning, logits often reflect the model’s confidence in its predictions. Higher logits indicate stronger confidence, resulting from more stable and consistent internal

Method	SPP-F F1	CCP F1	SLP Acc	CAS Acc	CAC Acc	Avg.
<i>Qwen2.5-7B-Instruct</i>						
Zero-shot	77.10	53.81	48.30	59.80	66.80	61.16
Legal Δ	83.14	68.53	64.80	64.80	75.80	71.41
w/o SFT	84.11	62.82	72.10	62.20	69.80	70.21
w/o GRPO	80.14	54.73	50.20	55.60	69.20	61.97
w/o Info-Gain	84.82	62.07	61.40	60.60	72.00	68.18
<i>Qwen2.5-14B-Instruct</i>						
Zero-shot	84.18	58.50	42.60	55.40	76.80	63.50
Legal Δ	85.56	62.79	67.10	72.40	79.40	73.45
w/o SFT	86.40	66.85	57.30	71.60	77.60	71.95
w/o GRPO	84.73	59.93	45.10	69.80	78.20	67.55
w/o Info-Gain	85.09	61.69	56.20	69.60	79.80	70.48

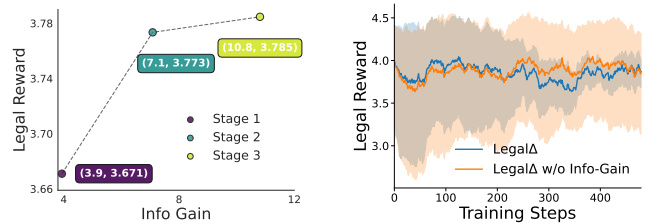
Table 2: Ablation Study on Legal Reasoning Tasks.



(a) Direct Response Scenario.

(b) CoT Prompting Scenario.

Figure 3: Logit Variations Induced by Information Gain Module During Training Process. We monitor the average logits values of Legal Δ models with and without information gain enhancement throughout the training process. In Figure 3(a), the model directly answers legal questions, while in Figure 3(b), the model generates step-by-step reasoning before providing answers.



(a) Correlation between Information Gain and Reward across Training Stages.

(b) Reward Progression Comparison during Training Process.

Figure 4: Analysis of Information Gain-Enhanced Reward Modeling Process. We collect legal rewards (LLM performance on legal tasks) and information gain during the training process. In Figure 4(a), we analyze the correlation between information gain and reward values across three distinct training stages. In Figure 4(b), we monitor and compare the reward progression of Legal Δ with and without information gain enhancement throughout the training process.

representations, which manifest as concentrated probabil-

ity distributions. To evaluate confidence improvement, we compare the logits performance of two variants of Legal Δ , depending on whether the information gain module is incorporated. As demonstrated in Figure 3, in direct response scenarios (Figure 3(a)), both models exhibit similar performance during early training stages. However, Legal Δ with information gain module progressively develops higher confidence levels, with increasingly pronounced performance gaps in later stages. This improvement becomes dramatically more evident in Chain-of-Thought prompting scenarios (Figure 3(b)), where Legal Δ demonstrates substantial and consistent logits gains throughout training while the baseline variant maintains relatively low confidence levels.

The improved logit values confirm that our information gain mechanism successfully identifies high-quality reasoning patterns. This is further supported by the positive correlation between information gain and legal reward, as illustrated in Figure 4(a), where both information gain and reward values demonstrate consistent growth across training stages. This correlation indicates that samples with higher information gain consistently receive higher rewards, showing that information gain provides another way to evaluate the quality of generated responses during GRPO training. Based on this positive correlation, our mechanism produces significant improvements in training efficiency and stability. As shown in Figure 4(b), Legal Δ with information gain enhancement maintains more stable training dynamics with substantially reduced reward variance compared to the baseline variant, creating more consistent reward signals aligned with reasoning quality.

5.4 Performance Quality Assessment

This experiment further explores the effectiveness of Legal Δ through performance quality assessment from both token-level and sequence-level perspectives.

Token-Level Analyses We first analyze the prominence of legal-related tokens. Legal token prominence, defined as the difference between logit values for specific legal tokens and the average logit value across the entire sequence, serves as an indicator of the model’s focus on legally relevant content.

As shown in Figure 5(a), we observe a progressive improvement across training stages: starting from the Zero-shot baseline with mainly negative prominence values, R1-Distill demonstrates initial improvements through high-quality reasoning data distilled from a large reasoning model, establishing foundational capabilities for identifying legally relevant content. Legal Δ further advances this progression through GRPO training, with the prominence values becoming notably higher and more samples achieving positive values compared to previous stages, indicating substantially enhanced attention to legally relevant tokens.

We further assess the quality of reasoning chains generated by our models through perplexity analysis for ground truth prediction. Specifically, we compare Legal Δ with and without information gain enhancement by asking each model to generate reasoning chains for questions and then feeding these reasoning chains along with questions to the vanilla LLM (Qwen2.5-32B-Instruct) to calculate the per-

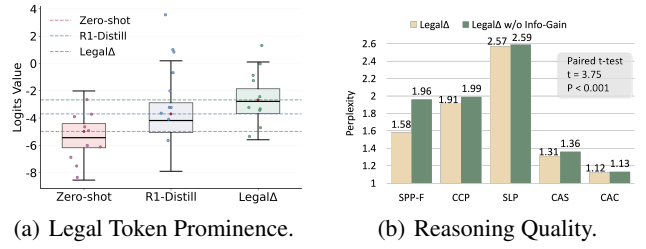


Figure 5: Token-Level Analysis. We analyze legal token prominence across different models using boxplots, where prominence indicates the model’s attention to legally relevant content. For reasoning quality evaluation, we compare Legal Δ with and without information gain module by assessing how well their generated chain-of-thought support answer prediction using LLM’s perplexity calculation.

Model	Short Reasoning		Long Reasoning	
	Length	Average Score	Length	Average Score
Legal Δ	222	77.41	299	73.69
Legal Δ w/o Info-Gain	251	76.36	341	69.10

Table 3: Response Length and Average Score Comparison of Legal Δ Models in Different Scenarios.

plexity of ground truth answers. Lower perplexity indicates that the reasoning chain better supports the correct answer prediction, reflecting higher reasoning quality. As demonstrated in Figure 5(b), Legal Δ with information gain enhancement consistently achieves lower perplexity values across legal reasoning tasks, with statistically significant improvements confirmed by paired t-test results. This demonstrates that the information gain module effectively enhances the quality of generated reasoning chains, enabling the LLM to generate more accurate responses.

Sequence-Level Analyses We further examine how the addition of information gain module affects reasoning quality across different scenario complexities. We categorize evaluation samples based on the inherent complexity of legal reasoning tasks, dividing them into short and long scenarios according to the response length generated by the baseline model Legal Δ w/o Info-Gain. As shown in Table 3, Legal Δ , which incorporates the information gain module consistently outperforms the baseline model across both short and long scenarios, generating more concise responses with superior performance scores. Notably, the performance improvement margin from adding the information gain module is more pronounced in long scenario, demonstrating that this incremental enhancement is particularly effective for legal reasoning tasks that require longer chain-of-thought.

6 Conclusions

This paper proposes Legal Δ , a novel framework that encourages the model to explore diverse reasoning trajectories and develop robust reasoning strategies by incorporating chain-of-thought-guided information gain into Reinforcement Learning with Verifiable Rewards (RLVR). Experi-

mental results demonstrate that Legal Δ effectively balances direct-answer and reasoning-enhanced generation modes, enabling the model to produce more concise and accurate responses. Further analysis shows that the reasoning processes generated by our method exhibit superior quality compared to baseline models, yielding significant performance gains.

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A Appendix

A.1 Dataset Statistics

As shown in Table 4, we use datasets from CAIL2018, JEC-QA, and LAIC2021 for training, covering five legal reasoning tasks. For evaluation, we conduct both in-domain evaluation using LawBench and LexEval benchmarks, and out-of-domain evaluation using six sub-tasks from the DiscLaw benchmark to assess model generalization capability.

A.2 Training Details

For Legal Δ , we apply LoRA to the query, key, value, output, gate, up, and down projection matrices. The maximum prompt length is set to 4096 tokens while maximum completion length is set to 512 tokens. We train for 1 epoch for the 14B model and 3 epochs for the other size models, using the cosine learning rate scheduler with a warmup ratio of 0.1. For optimization, we employ the 8-bit AdamW optimizer with $\beta_1 = 0.9$, $\beta_2 = 0.99$, and weight decay of 0.1. We set the maximum gradient norm to 0.1.

A.3 Legal Reward Construction

We design three distinct reward mechanisms based on the nature of expected outputs and evaluation requirements:

F1-Score Based Reward: Provides balanced evaluation for multi-label scenarios:

$$R_{F1} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (11)$$

This mechanism effectively penalizes both incomplete responses and excessive false positive predictions.

Distance-Based Reward: Offers continuous evaluation for numerical predictions:

$$R_{dist} = \exp \left(-\frac{|\log(\text{pred}) - \log(\text{true})|}{c} \right) \quad (12)$$

where c is a normalization constant set to 36 in our experiments. This design provides partial credit for predictions approximating the correct range.

Exact Match Reward: Provides strict binary evaluation for categorical selections:

$$R_{exact} = \begin{cases} 1 & \text{if prediction matches ground truth} \\ 0 & \text{otherwise} \end{cases} \quad (13)$$

These mechanisms are strategically applied: R_{F1} for criminal charge prediction and legal article identification, R_{exact} for financial amount calculation and multiple choice questions, and R_{dist} for sentencing prediction with wide value ranges.

A.4 Proof of Logits-Q Function Equivalence

Given dataset $\mathcal{D} = \{(q^i, a^i)\}_{i=1}^n$, the next-token prediction objective is:

$$\arg \max_{\pi} \sum_i \log \pi(a^i | q^i) \quad (14)$$

while the offline inverse reinforcement learning objective is:

$$\arg \max_Q \sum_i \left[Q(q^i, a^i) - \alpha \log \sum_a e^{Q(q^i, a)/\alpha} \right] \quad (15)$$

Task	Dataset	Metric	Quantity
Training			
Article Prediction	CAIL2018	F1	1,396 / 150
Cause Prediction	CAIL2018	F1	1,373 / 169
Penalty Prediction	CAIL2018	Distance	1,402 / 144
Case Analysis	JEC-QA	Accuracy	1,403 / 50
Financial Calculation	LAIC2021	Accuracy	1,407 / 50
In-domain Evaluation			
Article Prediction	CAIL2018	F1	500
Cause Prediction	CAIL2018	F1	500
Penalty Prediction	CAIL2018	Accuracy	1,000
Case Analysis	JEC-QA	Accuracy	500
Financial Calculation	LAIC2021	Accuracy	500
Out-of-domain Evaluation			
National Judicial Exam	NJE	Accuracy	537
Patent Agent Exam	PAE	Accuracy	118
Graduate Entrance Exam	UNGEE	Accuracy	320
Legal Basic Knowledge	LBK	Accuracy	275
Public Institutions Exam	PFE	Accuracy	170

Table 4: Dataset Statistics for Training and Evaluation.

Since $\pi_\theta(\cdot|q) = \text{softmax}(f_\theta(q, \cdot); \alpha)$, Eq. (14) can be rewritten as:

$$\arg \max_f \sum_i \left[f(q^i, a^i) - \alpha \log \sum_{a'} e^{f(q^i, a')/\alpha} \right] \quad (16)$$

Comparing Eq. (15) and Eq. (16), we obtain:

$$Q(q, a) = f(q, a) = \text{logit}_\theta(a|q) \quad (17)$$

A.5 Case Study

In Table 5, we compare the performance of Legal Δ and zero-shot models in analyzing a cross-border civil procedure case involving a commercial dispute between Chinese and German companies. The zero-shot model exhibits several critical limitations that significantly impair its practical utility. The model not only conflates fundamental procedural distinctions between domestic and international litigants—particularly by erroneously equating the 15-day appeal period for Chinese parties with the 30-day requirement for foreign-related cases—but also fails to recognize valid exceptions in litigation representation rules. These shortcomings are further exacerbated by the model’s generation of verbose and unfocused legal citations, which obscures crucial jurisdictional differences.

Legal Δ demonstrates markedly superior performance through its nuanced understanding of cross-border procedural law. The model accurately identifies and explains key asymmetries in legal treatment, including differentiated appeal periods and permissible representation forms. Its analytical approach represents a significant technical advancement, employing structured reasoning to strategically highlight legally determinative factors while maintaining clear and concise arguments. This sophisticated methodology stems from specialized training on party-specific rules and exceptions in international litigation. The model’s question-oriented output mechanism ensures direct relevance to the core legal issues, avoiding the digressive tendencies characteristic of zero-shot model responses. This demonstrates

that Legal Δ possesses stronger legal reasoning capabilities, enabling it to derive correct judgments and conclusions through key facts.

A.6 Prompt Template

In this section, we introduce the prompt used in Legal Δ . Figure 6 presents the prompt used in direct mode. Figure 7 indicates the prompt used with Chain of Thought during the rollout process.

Case Analysis Task on Cross-border Civil Procedure	
Criminal Fact	Dayang Company, with its domicile in District A, City A of China, signed a sales contract with Kerrys Company in Germany. The German Kerrys Company has an office established in China, and later the parties had a dispute over quality issues. Regarding statements about transnational civil litigation, which of the following options is INCORRECT? A: Both parties may agree to determine the court with jurisdiction over the dispute B: The court may serve litigation documents to Kerrys Company's office in China C: Both parties have an appeal period of 30 days D: Kerrys Company may appoint a lawyer from its home country to serve as a litigation representative in a non-lawyer capacity
Vanilla	<reasoning> Option A Analysis: According to Article 34 of the Civil Procedure Law of the People's Republic of China, parties to contract or other property rights disputes may agree in writing to choose the people's court with jurisdiction at the defendant's domicile, contract performance location, contract signing location, plaintiff's domicile, or location of the subject matter that has actual connection with the dispute. Therefore, both parties can agree to determine the court with jurisdiction over the dispute. Option B Analysis: According to Article 265 of the Civil Procedure Law of the People's Republic of China, in transnational civil litigation, if the law of the country where the recipient is located allows service by mail, service by mail is permitted. If no service receipt is returned within three months from the date of mailing, but various circumstances sufficiently indicate delivery has occurred, it shall be deemed served upon expiration of the period. Since Kerrys Company has an office in China, the court can serve litigation documents to Kerrys Company's office. Option C Analysis: According to Article 267 of the Civil Procedure Law of the People's Republic of China, in transnational civil litigation, the appeal period is 30 days. Therefore, both parties have an appeal period of 30 days. Option D Analysis: According to Article 298 of the Civil Procedure Law of the People's Republic of China, when foreigners, stateless persons, foreign enterprises and organizations file lawsuits or respond to lawsuits in people's courts and need to appoint lawyers for litigation representation, they must appoint lawyers from the People's Republic of China. Therefore, Kerrys Company cannot appoint a lawyer from its home country to serve as a litigation representative in a non-lawyer capacity. Reasons for Excluding Options: Options A, B, and C all comply with legal provisions, while Option D does not comply with legal provisions. Correct Option Confirmation: Option D does not comply with legal provisions, therefore it is the incorrect option. </reasoning> <answer> D </answer>
LegalΔ	<reasoning> Option A Analysis: Correct, both parties may agree to determine the court with jurisdiction. Option B Analysis: Correct, the court may serve documents to foreign parties at their offices in China. Option C Analysis: Incorrect, Chinese parties have an appeal period of 15 days, foreign parties have 30 days. Option D Analysis: Correct, foreign parties may appoint lawyers from their home country to serve in a non-lawyer capacity as representatives. Reasons for Excluding Options: Option C - Chinese citizens have 15 days, foreign citizens have 30 days. Correct Option Confirmation: C </reasoning> <answer> C </answer>

Table 5: Case Study

Prompt for Direct Mode

[SYSTEM]: User-assistant dialogue. The user asks questions and the assistant solves them. The answer is contained in <answer> </answer> tags.

[USER]: You are a legal expert, answer based on **requirements** below:

****Task Description:****

Please simulate a judge to determine charges based on the facts below.

****Input:****

Case fact description

****Output Format:****

Please output your judgment result in the following format:

<answer>

[charges]{specific charges}<eo>

</answer>

- `{specific charges}`: String representing the specific charges you determined based on the case facts, multiple charges separated by commas, charges should not appear in "XX crime" format (like "theft crime"), but expressions containing the word "crime" are allowed.
- `[charges]`: Indicates the start flag of the answer output.
- ``: Indicates the end flag of the output.

Below is the specific case description:

Facts: The Shangshui County People's Procuratorate alleges: The defendant...

[ASSISTANT]:

<answer>

[charges]refusal to execute judgment<eo>

</answer>

Figure 6: Prompt Template of Direct Mode.

Prompt for Reasoning Mode

[SYSTEM]: User-assistant dialogue. The user asks questions and the assistant solves them. The assistant first thinks through the reasoning process, then provides the user with an answer. The reasoning process and answer are contained in <reasoning> </reasoning> and <answer> </answer> tags respectively.

[USER]: You are a legal expert, answer based on **requirements** below:

****Task Description:****

Please simulate a judge to determine charges based on the facts below.

****Input:****

Case fact description

****Output Format:****

Please output your judgment result in the following format:

<reasoning>

detailed analysis process

</reasoning>

<answer>

[charges]{specific charges}<eo>

</answer>

- `{specific charges}`: String representing the specific charges you determined based on the case facts, multiple charges separated by commas, charges should not appear in "XX crime" format (like "theft crime"), but expressions containing the word "crime" are allowed.
- `[charges]`: Indicates the start flag of the answer output.
- `<eo>`: Indicates the end flag of the output.

Below is the specific case description:

Facts: The Shangshui County People's Procuratorate alleges: The defendant...

Figure 7: Prompt Template of Reasoning Mode.